

Tucker L. Capps

Curriculum Vitae

capps@ou.edu | LinkedIn | Research Website | GitHub

Education

The University of Oklahoma, Norman, Oklahoma

Doctor of Philosophy, Astrophysics, **Qualifying Exams Passed:** Jan 2025

May 2029

Master of Science, Data Science and Analytics, **GPA:** 4.0

August 2025

Bachelor of Science, Astrophysics, *Special Distinction*

May 2023

Bachelor of Arts, Mathematics, *Special Distinction*

May 2023

Research Interests

- **Secular Evolution of Galactic Disks**

- Radial migration, angular momentum transport, and the response of disks to minor mergers
- Spiral structure: transient corotating patterns, swing amplification, and tidally induced waves

- **Kinetic Theory of Stellar Disks**

- Quasilinear and stochastic descriptions of collisionless relaxation
- Designing controlled simulations as direct tests of analytic dynamical theory

- **High-Performance and Numerical Astrophysics**

- Large N-body and AMR simulation campaigns, parallel computation, data-intensive workflows

Research Experience

Doctoral Research, Dr. Michael R. Hayden

Aug 2024 – Present

The University of Oklahoma

Radial Migration and the Secular Evolution of Galactic Disks

- Lead a controlled N-body study of how a minor merger alters secular evolution in a Milky Way-like disk, using matched isolated and perturbed runs of 50 million disk and 20 million halo particles.
- Establish that the encounter acts as a time rescaling of angular momentum redistribution, with a single acceleration factor setting the evolutionary clock of the perturbed disk.
- Construct equilibrium initial conditions from action-based distribution functions (AGAMA) and evolve them with the AMR code RAMSES on HPC clusters.
- Measure spiral amplitudes, pattern speeds, and mode coherence times from Fourier decompositions of the disk, and use action-space diagnostics to constrain the transport mechanism.
- Develop the full Python analysis stack for phase-space diagnostics, spectrograms, and migration statistics.
- First-author manuscript in preparation. Results presented as a contributed talk at the 57th Meeting of the Division on Dynamical Astronomy.

Master's Thesis, Dr. Michael R. Hayden

Aug 2024 – Jul 2025

The University of Oklahoma

- Developed machine-learning framework (XGBoost) to predict stellar ages from large spectroscopic surveys ($\sim 400\text{k}$ stars).
- Implemented feature selection, cross-validation, and hyperparameter optimization.
- Combined Bayesian isochrone fitting with data-driven modeling on HPC resources.

NASA OSTEM Intern, Dr. Irina Kitiashvili

Jun 2024 – Aug 2024

NASA Ames Research Center

- Computed acoustic power maps of solar active regions using large observational datasets.
- Developed random-forest model to predict magnetic field strength from helioseismic diagnostics.
- Worked on the Pleiades supercomputer and presented results to research staff.

Undergraduate Research, Dr. Xinyu Dai

2021 – 2023

The University of Oklahoma

- Analyzed microlensing magnification maps using HPC simulations and Python-based image processing.

Publications

Capps, T. L., Hayden, M. R., Tepper-García, T., Bland-Hawthorn, J., “Satellite-driven acceleration of secular evolution in a Milky Way–like disk,” *in preparation*.

Talks and Presentations

Capps, T. L. et al., “Satellite-driven acceleration of radial migration in a Milky Way–like disk,” 57th Meeting of the Division on Dynamical Astronomy, 2026 (Talk).

Capps, T. L. et al., Poster Presentation, 247th Meeting of the American Astronomical Society, 2025.

Advanced Training

RAMSES New User School, Observatoire astronomique de Strasbourg, France

Oct 2026

Teaching Experience

Graduate Teaching Assistant

Aug 2024 – Present

The University of Oklahoma

- Teach undergraduate astronomy labs and telescope observing techniques.
- Grader for ASTR 4303, Stellar Astrophysics, and Demonstration Room Assistant for the introductory physics sequence.
- Mentor students in scientific computing, data analysis, and research communication.
- Contributed to the revision of laboratory curriculum materials.

Peer Learning Assistant

2021 – 2023

The University of Oklahoma

- Tutored introductory physics courses and delivered exam-review lectures.
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Community Engagement

Imaging Lead and Engineer	2024 – Present
Lunar Sooners Outreach	
• Support public observing events and maintain astronomical instrumentation. • Deliver STEM outreach programming to K–12 audiences, reaching more than 2000 students since Fall 2025. • Produce astrophotography and Manim animations for public science communication. • Interviewed as a departmental source for “Saving the Night Sky,” <i>Oklahoma Magazine</i>, July 2026.	

Honors and Awards

Robberson Research Travel Grant, \$1,000	Spring 2026
Avenir Research Fellowship, ~\$25,000	Spring - Summer 2026
Marvin Martin Mueller Scholarship, \$2,000	2022–2023
Michael Ruby History of Physics Scholarship, \$2,000	2021–2022
Meritorious Scholarship in Physics and Astronomy	2021–2023

Technical Skills

Programming:	Python, C++, SQL, L ^A T _E X
Astrophysical Software:	RAMSES, AGAMA, pynbody
High Performance Computing:	MPI parallel computing, HPC clusters
Numerical Methods:	N-body dynamics, adaptive mesh refinement, action-angle methods, Fourier and spectral analysis
Machine Learning:	XGBoost, scikit-learn
Science Communication:	Manim, astrophotography and image processing

References Available Upon Request